

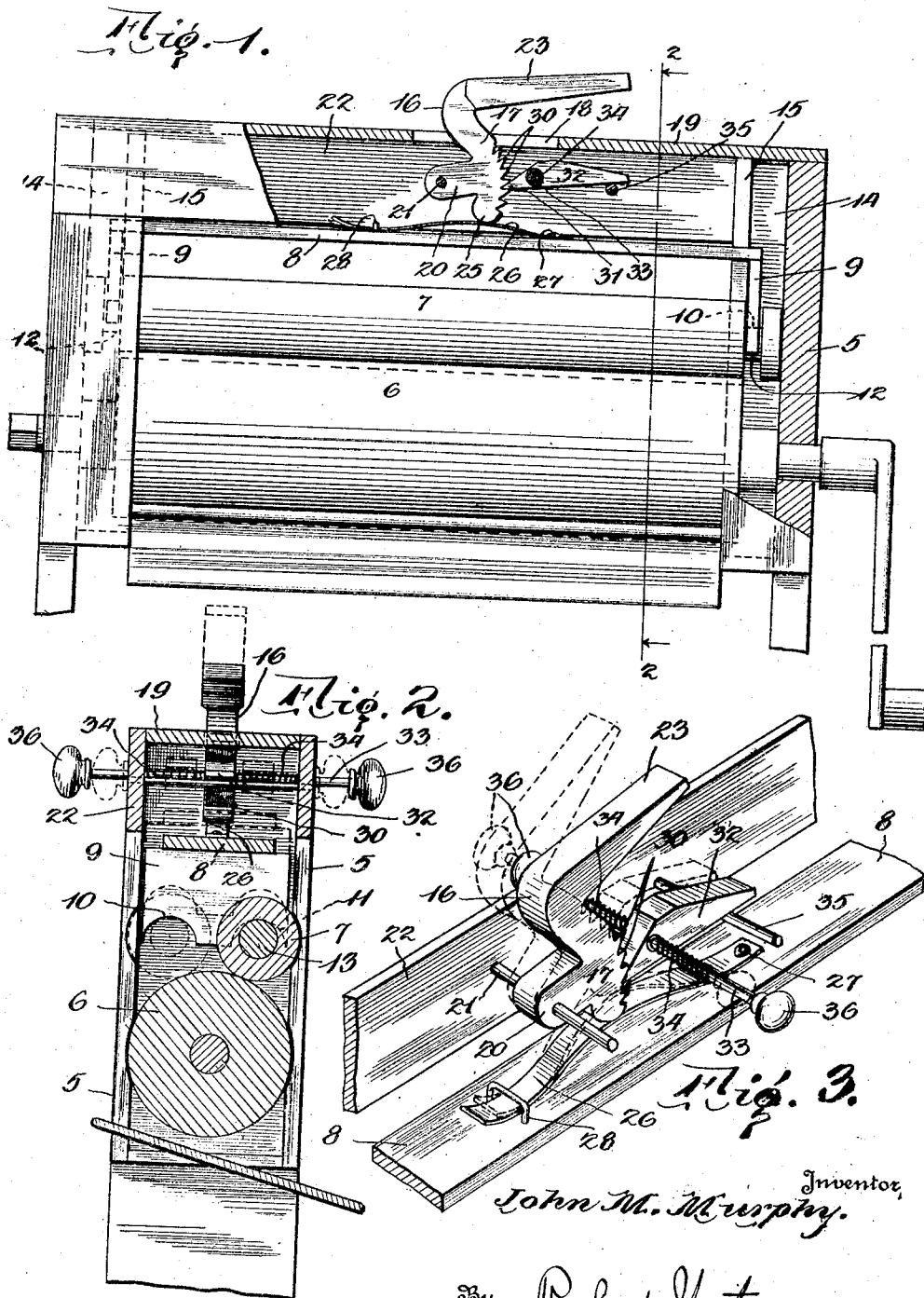
Jan. 27, 1925.

J. M. MURPHY

1,524,524

WRINGER

Filed Aug. 6, 1924



UNITED STATES PATENT OFFICE.

JOHN M. MURPHY, OF SHAMOKIN, PENNSYLVANIA, ASSIGNOR OF ONE-HALF TO
CHARLES K. MORGANROTH, OF SHAMOKIN, PENNSYLVANIA.

WRINGER.

Application filed August 6, 1924. Serial No. 730,403.

To all whom it may concern:

Be it known that I, JOHN M. MURPHY, a citizen of the United States, residing at Shamokin, in the county of Northumberland and State of Pennsylvania, have invented certain new and useful Improvements in Wringers, of which the following is a specification.

This invention relates to safety pressure releases for wringers, and has for one of its objects to provide a device of this character which will be simple in construction, comparatively inexpensive to manufacture and more efficient in use than those which have been heretofore proposed.

In the power operated wringers which are now commonly employed in connection with electric washing machines, it is customary to provide a pair of wringer rolls, one of which is adjustable to and from the other in order to control the pressure exerted between the rolls, which pressure, in common practice, is quite considerable. Many serious accidents have occurred in the use of such wringers due to the operator inadvertently placing his hand in such position that it is gripped by the wringer rolls and drawn between them, thereby crushing or tearing the hand of the operator to a serious extent.

Various devices have been proposed for preventing the hand of the operator from thus being drawn into the wringer and for releasing the pressure of the rolls in cases of such emergency, and it is to the latter class of device that the present invention relates.

In all the prior devices of this character of which I am familiar, the pressure of the releasing mechanism is provided with a trip mechanism usually having a lever or handle extending from the top of the wringer, which may be manually operated to release one of the rolls in case of the operator's hand being caught between the rolls, but such devices do not take into consideration the natural tendency of the operator to attempt to withdraw his hand which is caught by placing the other hand against the front of the wringer and pulling backward on the hand which is caught. The present invention, on the other hand, provides a mechanism having a trip extending transversely

of the wringer and which may be operated from either side thereof, but which is located in the most natural position for the operator to strike in attempting to withdraw the hand which is caught from the rolls.

With the above and other objects in view, which will appear as the description proceeds, the invention consists in the novel details of construction and combinations and arrangements of parts more fully hereinafter described and particularly pointed out in the appended claims.

Referring to the accompanying drawings forming a part of this specification, in which like reference characters designate like parts in all the views:

Figure 1 is a front elevational view of a wringer provided with a safety pressure release mechanism constructed in accordance with the present invention;

Fig. 2 is a vertical transverse sectional view taken approximately on the plane indicated by the line 2—2 of Fig. 1, looking in the direction of the arrows; and,

Fig. 3 is an enlarged perspective view of the trip mechanism shown in Figs. 1 and 2.

Referring more especially to the said drawings the numeral 5 indicates a frame or casing of the wringer in which is suitably journaled a lower roll 6 and an upper roll 7. The lower roll 6, which is preferably somewhat larger in diameter than the upper roll 7, is journaled against horizontal or vertical movements, while the upper roll 7 is so constructed and arranged that it may be released to move either upwardly or transversely of the wringer frame 5. This mounting of the upper roll is preferably accomplished by means of a bridge member 8, which extends above and parallel to the axis of the roll 7, and which is provided at its ends with the downwardly extending ears 9, the lower faces of which are provided with the transversely spaced substantially semi-circular bearing recesses, 10 and 11, as will be clear from Fig. 2.

These recesses are adapted to receive the ends 12 of the spindle or axle 13, upon which the roll 7 is mounted. It is to be understood that the roll may be shifted from left to right, and vice versa, as indicated in dotted lines in Fig. 2, in order to provide for feed-

ing the clothes from either side, as may be desired without the necessity of reversing the entire wringer.

The ears 9 are of substantially the same width as the interior of the casing 5 and are slidably mounted in recesses 14 formed in the end portions of the wringer frame 5, the edges of said recesses being provided with suitable projections or guides 15, for preventing the longitudinal displacement of the bridge member, while at the time permitting of its vertical movement.

In order to provide for the controlling of the pressure which may be exerted between the rolls 6 and 7, there is provided a substantially L-shaped lever 16, having a leg 17 extending through an opening 18, formed in the cover plate 19 of the frame 5, the leg being provided with a substantially right angular extension 20 through which passes a pivot bolt or shaft 21, the ends of which project through the side members 22 of the frame 5. The L-shaped lever 16 is also provided with the leg 23, which projects in a substantially horizontal direction above the cover plate 19, whereby the lever may be manually operated to control the pressure of the roll 7 upon the roll 6, as will presently appear.

The leg 17 of the lever 16 is preferably provided with a rounded end 25 which normally rests upon and engages a leaf spring 26, one end of which is rigidly secured to the upper surface of the bridge member 8, as at 27, while the other end of the spring projects through a staple or other guiding member 28, also carried by the bridge member 8. This end of the spring is slidably mounted relative to the guiding member 28 so that it is free to move therethrough, as will be readily understood.

In order to retain the lever 16 and the bridge member 8 in any desired adjusted position to secure a desired pressure between the rolls, the leg 17 of the lever 16 is provided with a plurality of notches or serrations 30, which are adapted to be engaged by the pointed toe 31 of a dog 32, which is pivotally mounted upon a transversely extending rod or shaft 33 in co-operative relation with the lever 16. The dog 32 is rigidly attached to the pivot member 33 and this member is preferably slidably mounted in the side members 22 of the frame 5 so that the rod, together with the dog 32, may be transversely moved relative to the wringer frame and lever 16 to disengage its toe 31 from the serrations 30. Suitable coiled springs 34 surround the rod 33 upon each side of the dog 32 and bear against the side members 22 whereby the dog will normally be maintained in a substantially central position within the wringer frame with its toe 31 in alignment with the leg 17 of the lever 16.

It will be noted that the dog 32 is pivoted closer to one end than to the other, whereby it will be subject to the action of gravity which will tend to turn the same about the axis of the pivot 33 in a clockwise direction, as viewed in Fig. 1. A suitable stop pin 35 is mounted in the side members 22 upon which the free end of the dog may rest to prevent it from assuming the position where its toe 31 will not engage with the serrations 30 of the lever 16.

The pivot rod or shaft 33 which carries the dog, as above stated, is slidably mounted in the side members 22 and projects some distance beyond the said members, each end of the said dog being provided with a knob or fingerpiece 36, as is clearly shown in Figs. 2 and 3.

The operation of the device will be clear from the foregoing but it may be briefly summarized as follows: The wringer being placed in the proper position with the upper roll 17 occupying either the right or left hand recesses in the ears 9, the pressure that the roll may exert upon the lower roll 6 is adjusted to the desired point by merely pressing down upon the leg 23 of the lever 16. This will rock the lever about its pivot 21 to cause the rounded end 25 of the leg 17 to press upon the leaf spring 26, which pressure will in turn be transmitted from the said spring to the bridge member 8 and through its ears 9 to the ends 12 of the roll shaft 13. The parts are now in a position for use and the clothes may be fed between the rolls in the usual and well known manner.

Should the operator accidentally catch his hand or a portion of his clothing between the rolls so that the latter begin to draw the caught member between them, as above stated, his most natural inclination would be to place his free hand against the forward member 22 of the wringer frame and to endeavor to pull the other hand out of the rolls. In so doing however, with the present invention, the free hand will automatically strike the knob 36 on the end of the pivot rod 33 which, being slidably mounted in the side members 22, may be pushed inwardly against the compression of one of the springs 34 and carry with it during its transverse movement, the dog 32. As soon as the dog 32 clears the side of the leg 17 of the lever 16, the latter will, of course, be released and will move in a counterclockwise direction about its pivot 21, thereby releasing the pressure upon the spring 26 and the bridge member 8, so that the roll 7 will be released and the caught hand or clothing be readily withdrawn before serious injury or damage has been sustained. Of course, as soon as the knob 36 is released, the spring 34 which has been compressed will return the pivot rod 33 and the dog 32 to

its medial position, in alignment with the leg 17 of the lever 16, so that by again compressing the leg 23 of the lever, the parts may be re-engaged and the desired pressure secured between the rolls.

While one form of the invention has been illustrated and described it is obvious that those skilled in the art may vary the details of construction as well as the arrangement of parts without departing from the spirit of the invention, and therefore it is not wished to be limited to the above disclosure except as may be required by the accompanying claims.

What is claimed is:

1. A safety pressure release for roll wringers comprising means for adjusting one of the wringer rolls to control the pressure; means for locking said adjusting means in a desired position, said locking means being laterally reciprocable in either direction to disengage said adjusting means; and means for yieldingly holding said locking means in engagement with said adjusting means.

2. A safety pressure release for roll wringers comprising means for yieldingly forcing one of the rolls toward the companion roll; means co-acting with said forcing means to retain it in a desired position, said retaining means being laterally movable in either direction to disengage said forcing means to release the same; and resilient means for normally maintaining said retaining means in engagement with said forcing means.

3. A safety pressure release for roll wringers comprising a lever for forcing one of the rolls toward its companion roll; and a locking dog normally engaging said lever to retain it in a set position, said dog being laterally movable to disengage said lever.

4. A safety pressure release for roll wringers comprising a pivoted lever for forcing one of the rolls toward its companion roll, said lever having a serrated portion; and a locking dog normally engaging the serrations of said lever to retain it in a set position, said dog being laterally movable to disengage said lever serrations.

5. A safety pressure release for roll wringers comprising a pivoted lever for forcing one of the rolls toward its companion roll, said lever having a serrated portion; a pivoted locking dog normally engaging the serrations of said lever to retain it in a set position, said dog being laterally movable to disengage said lever serrations; and a stop for limiting the pivotal movement of said dog in one direction.

6. A safety pressure release for roll wringers comprising a pivoted lever for forcing one of the rolls toward its com-

panion roll; a locking dog adapted to engage said lever to retain it in a set position, said dog being laterally movable out of alignment with said lever to release the same; and means for normally holding said dog in alignment with said lever.

7. A safety pressure release for roll wringers comprising a pivoted lever for forcing one of the rolls toward its companion roll, said lever having a serrated portion; a locking dog adapted to engage the serrations of said lever to retain it in a set position, said dog being laterally movable out of alignment with said lever to release the same; and yielding means for normally holding said dog in alignment with said lever.

8. A safety pressure release for roll wringers comprising a pivoted lever for forcing one of the rolls toward its companion roll, said lever having a serrated portion; a pivoted locking dog adapted to engage the serrations of said lever to retain it in a set position, said dog being laterally movable out of alignment with said lever to release the same; yielding resilient means for normally holding said dog in alignment with said lever; and a stop for limiting the pivotal movement of said dog in one direction.

9. The combination with a wringer comprising a frame, a pair of rolls mounted therein, one of which is movable toward and from the other to vary the pressure between them, and a bridge member for moving said movable roll toward its companion roll, of a lever pivotally mounted in said frame, provided with a portion adapted to urge said bridge member and movable roll toward said companion roll; and a dog mounted in said frame normally engaging said lever to maintain it in set position, said dog being also laterally movable in said frame to disengage said lever.

10. The combination with a wringer comprising a frame, a pair of rolls mounted therein, one of which is movable toward and from the other to vary the pressure between them, and a bridge member for moving said movable roll toward its companion roll, of a lever pivotally mounted in said frame, provided with a portion adapted to urge said bridge member and movable roll toward said companion roll, said lever also having a serrated portion; a dog pivotally mounted in said frame normally engaging the serrations of said lever to maintain it in set position, said dog being also laterally movable in said frame to disengage said lever; and spring means for normally maintaining said dog in alignment with said lever.

11. The combination with a wringer comprising a frame, a pair of rolls mounted therein, one of which is movable toward

and from the other to vary the pressure between them, and a bridge member for moving said movable roll toward its companion roll, of a substantially L-shaped lever pivotally mounted in said frame, provided with a portion adapted to urge said bridge member and movable roll toward said companion roll, said lever also having a serrated portion; a dog pivotally mounted in said frame normally engaging the serrations of said lever to maintain it in set position, said dog being also laterally movable in said frame to disengage said lever; spring means for normally maintaining said dog in alignment with said lever; and a stop pin for limiting pivotal movement of said dog in one direction.

In testimony whereof I hereunto affix my signature.

JOHN M. MURPHY.